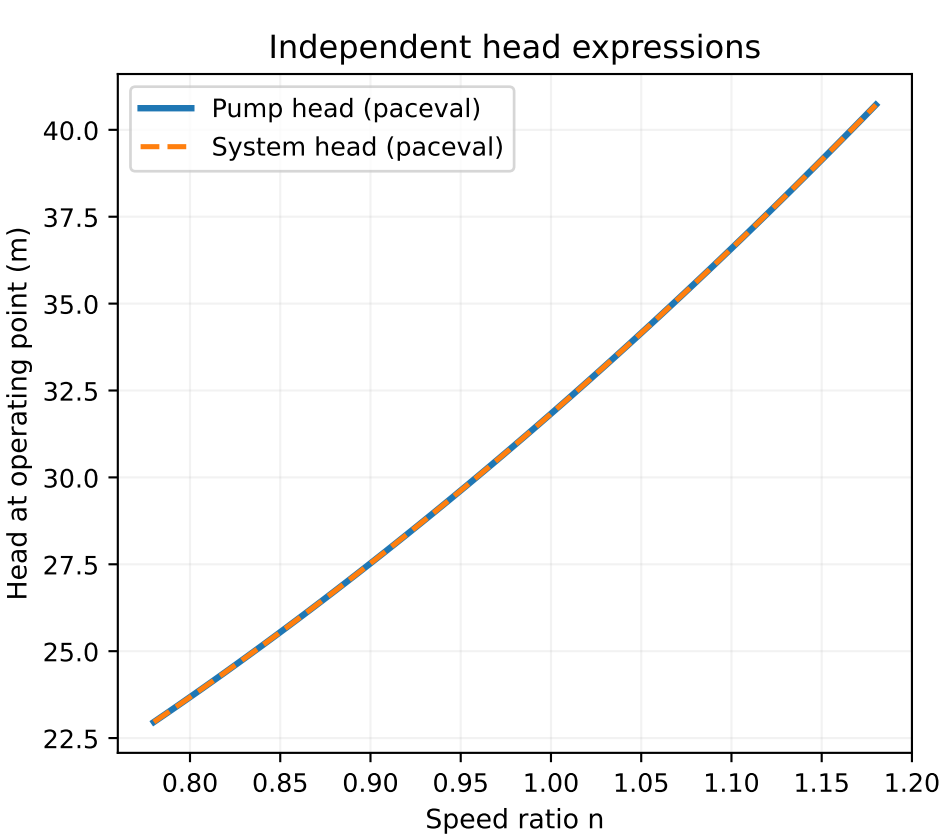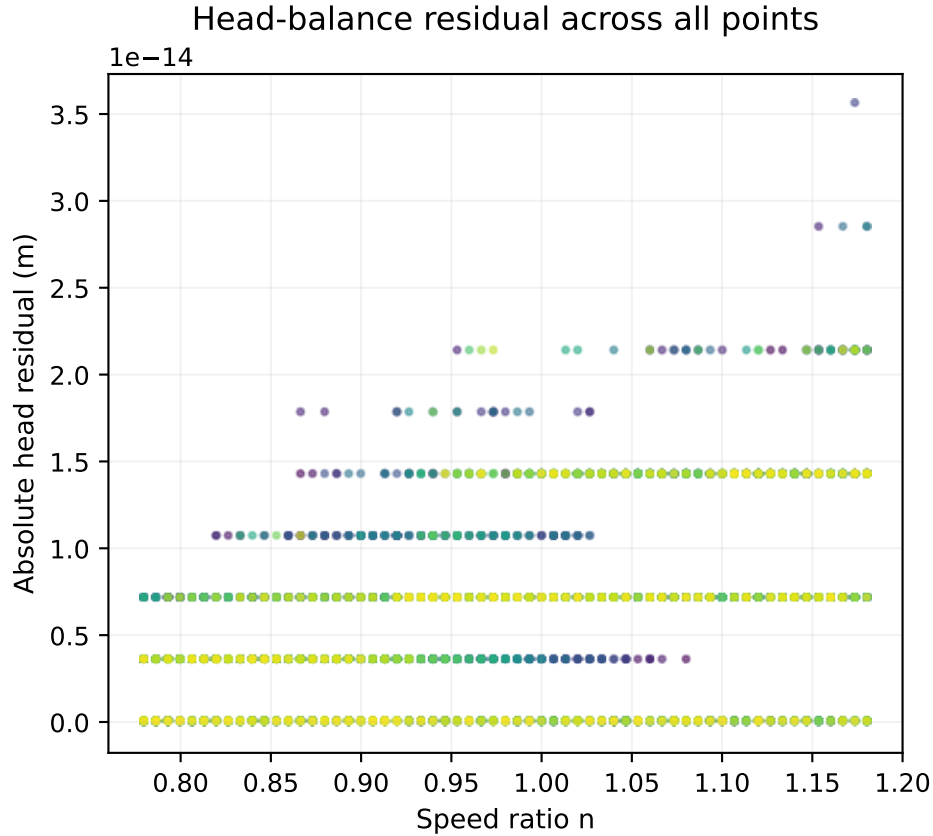Operating-point consistency: pump and network heads coincide